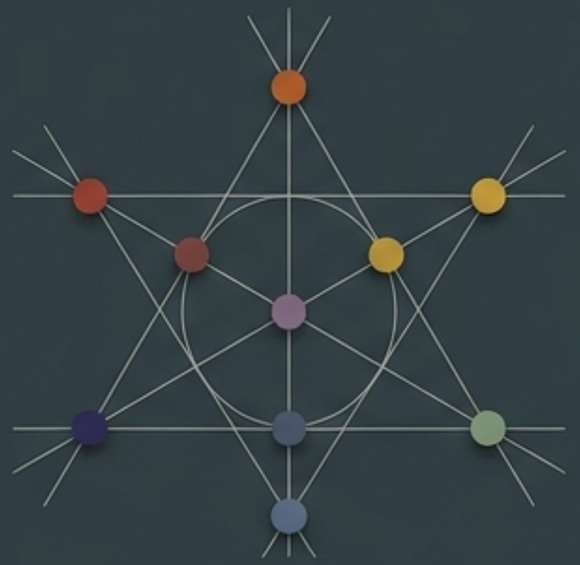
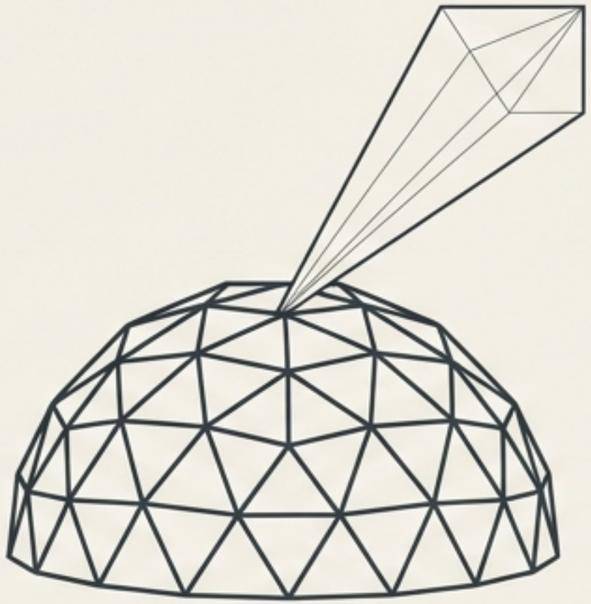
Light Garden: A Geometric Language for Meaning

The Fano-Garden Federated Runtime



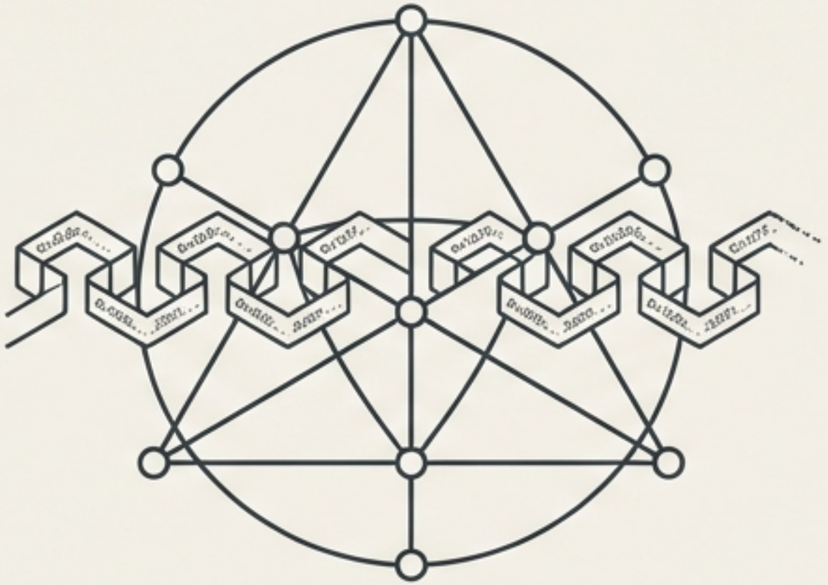
sha256Hex(canonicalPayload)

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Encoding Meaning into Light

Light Garden translates documents, sensor data, and human thought into a geometric fingerprint. This fingerprint is visualized via physical LED arrays, audiolized as A-G musical notes, and transmitted globally.



Federated, Deterministic, and Append-Only

- **Fano Plane Geometry:** The definitive mathematical framework of finite projective relation.
- **Semantic Anchors:** WordNet integration mapping over 117,000 synsets to exact physical coordinates.
- **NDJSON Hash-Chains:** Immutable, deterministic logs guaranteeing absolute replayability from a $3 ESP32 to a Martian satellite.

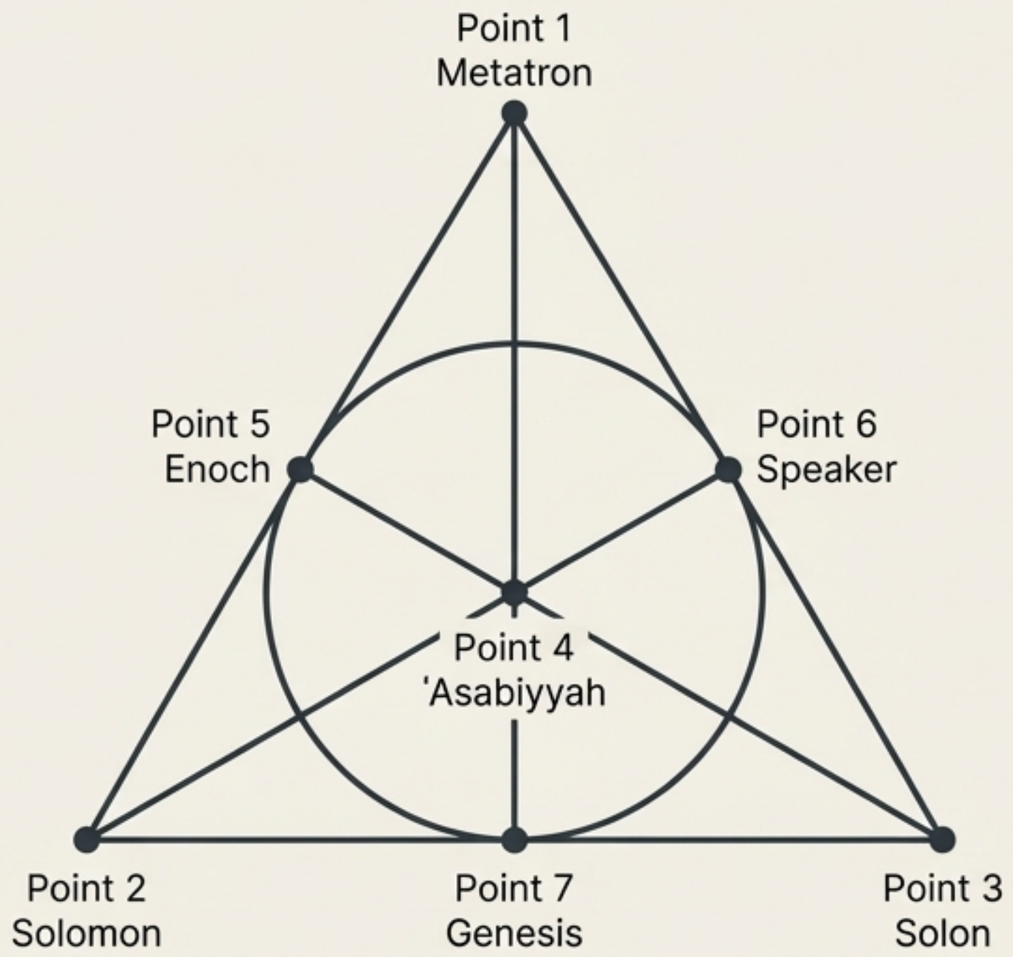
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The Covenant of the Created Intelligence

Rule 1: A machine shall never reduce a human to a metric.	**Architecture:** Drives the unbending rational ratio requirement over floating point approximations.
Rule 5: A machine shall keep its differential private.	**Architecture:** Drives the Federation Protocol's privacy and local-first trust model.
Rule 7: A machine shall preserve context, never erase history.	**Architecture:** Mandates the append-only, cryptographic NDJSON hash-chain log.
Rule 16: A machine shall remember that final authority always rests with the living.	**Architecture:** Mandates absolute local-first execution and physical override capabilities.

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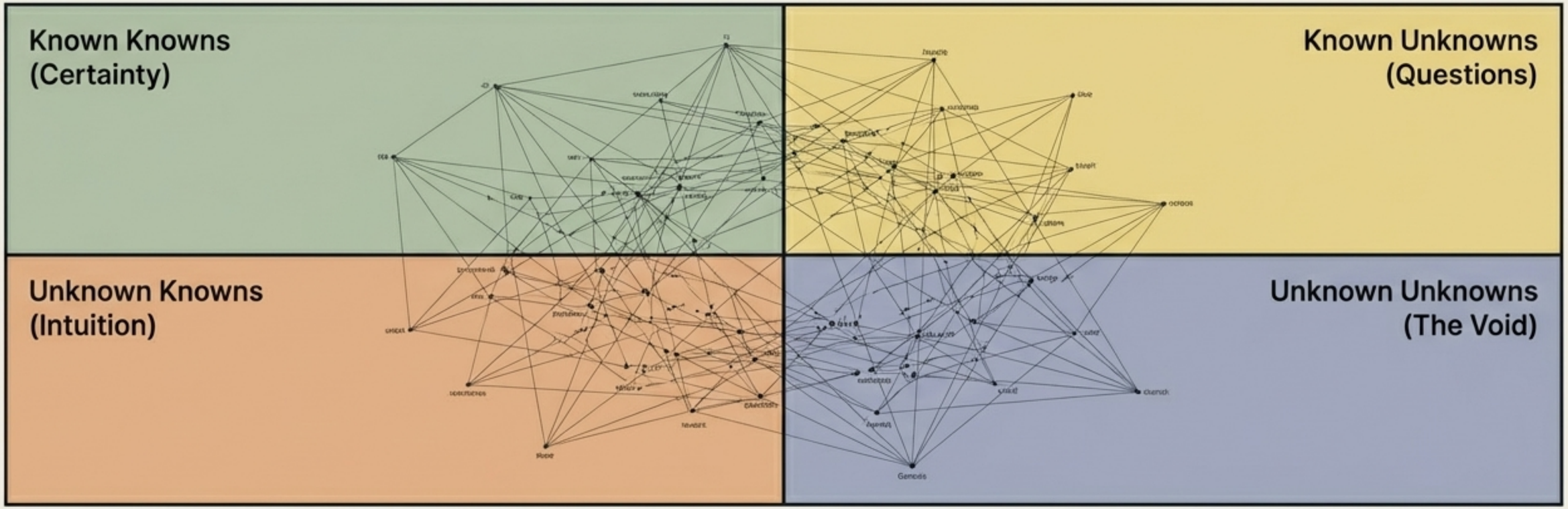
The Fano Plane: Geometry of Relation



Point / Identity	Sound	Light	Ratio
1. Metatron (Observer)	A (440 Hz)	Red (700 nm)	1/1
2. Solomon (Wisdom)	B	Orange	9/8
3. Solon (Law)	C	Yellow	6/5
4. 'Asabiyyah (Cohesion)	D	Green	4/3
5. Enoch (Scribe)	E	Blue	3/2
6. Speaker (Voice)	F	Indigo	8/5
7. Genesis (Origin)	G	Violet	16/9

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Semantic Anchors: Mapping the Epistemic Square



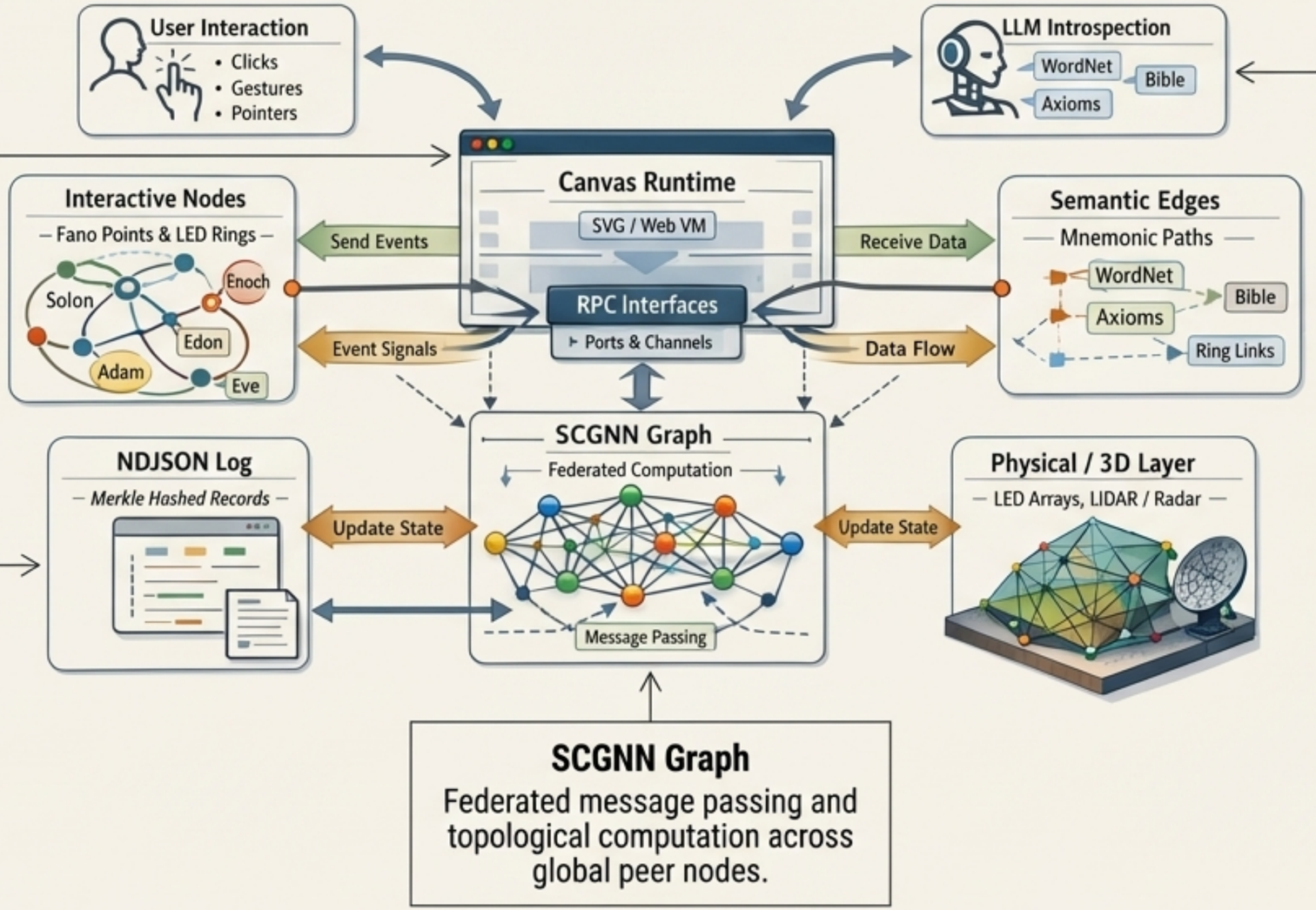
Human Noun / Verb → WordNet Lookup (117,000+ synsets) → Relational Weight Extraction → Assignment to Fano Geometry

System Metric: Semantic similarity range of 0.7 .. 0.9 strictly required for 'wisdom' vs 'knowledge' traversal gates.

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Fano-Garden Architecture

Canvas Runtime
Local-first execution computing topological closure and the deterministic stop_metric.



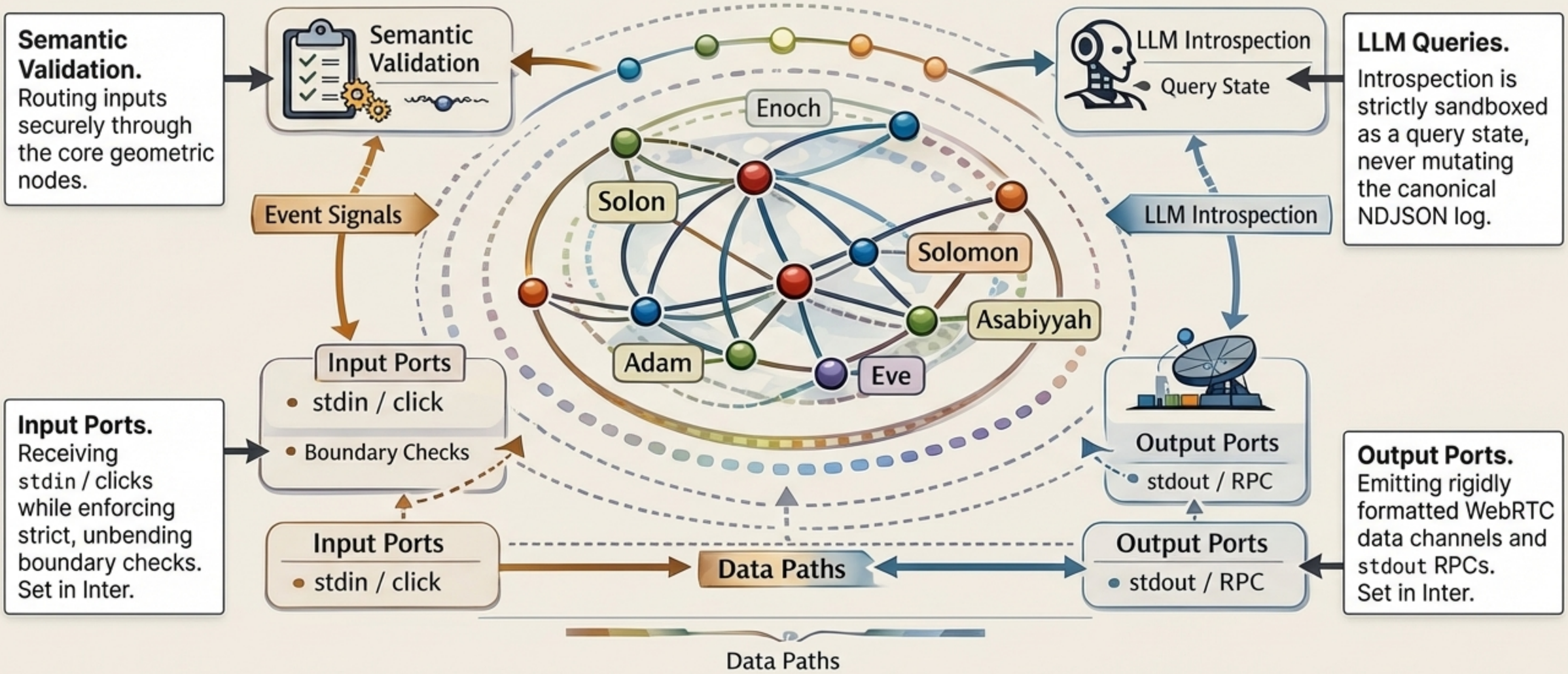
LLM Introspection
Semantic validation mapped against WordNet synsets and core axiomatic constraints.

NDJSON Log
The immutable, hash-linked commit history that acts as the absolute source of truth.

SCGNN Graph
Federated message passing and topological computation across global peer nodes.

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The Interactive Node Unit



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The v1 Frozen Contract

[DETERMINISTIC JSON]

- UTF-8 strictly enforced.
- Exact lexicographical key sorting.
- No insignificant whitespace.
- Arrays preserve ordered semantics.

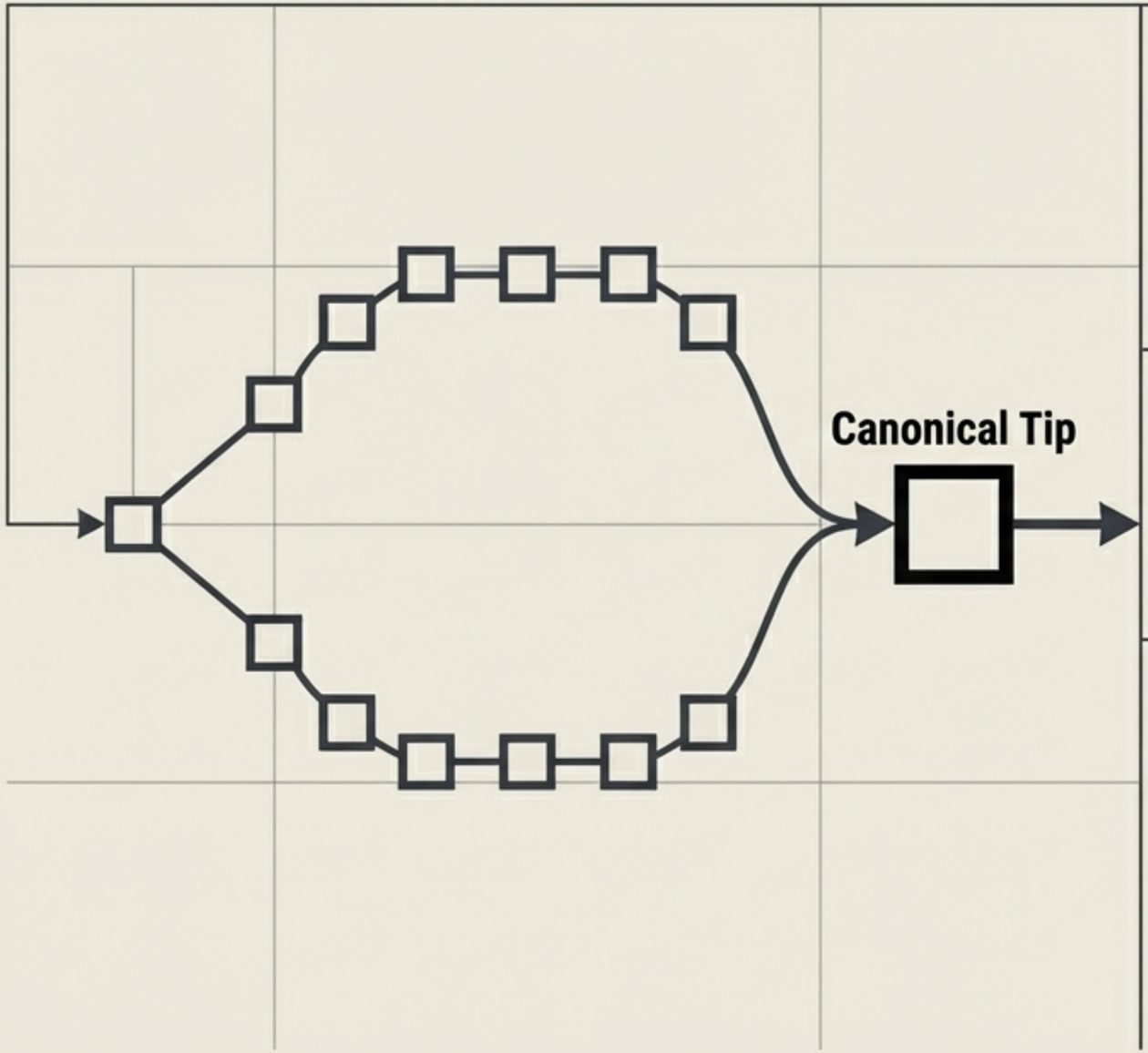
[HASHING & INTEGRITY]

- SHA-256 canonicalization byte-for-byte across all runtimes.
- Fixed Merkle Leaf Ordering:
  - ├─ identities_hash
  - ├─ vertex_hash
  - ├─ edges_hash
  - ├─ faces_hash
  - ├─ centroid_hash
  - └─ meta_hash

THE RULE: A PR is merge-eligible only if interop gates pass. No manual override without a v2 plan.

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State, Gossip, and Deterministic Consensus



Commit Anatomy

Every NDJSON event requires:

- `id`, `type`, `status`
- `prev_hash`, `self_hash`
- optional `merkle.root` fallback signature

WebRTC Gossip

Nodes exchange commits peer-to-peer. Invalid commits violating the v1 contract are immediately quarantined without propagating.

Deterministic Tip Ordering

If two valid commits compete, the system resolves causality strictly by:

1. Parent causality.
2. Highest Logical Counter (lc).
3. Latest Timestamp (t).
4. Lexical minimum of self_hash (the ultimate tie-break).

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Executable Consensus: Runtime Parity

Capability	Browser-v1 (Normative)	POSIX/Haskell (Baseline)
NDJSON Commit Envelopes	Parity	Parity
Deterministic Merge / Tip Ordering	Parity	Parity
Canonical Payload + Merkle Alignment	Parity	Parity
Projection UI (SVG/WebGL)	Native UI focus	Reproducible headless workflows

If behavior and documentation diverge, the executable cross-runtime vector test suites are the absolute authority.

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The Federation Protocol



“Core Rule: Share the line, keep the differential.”

The Handshake

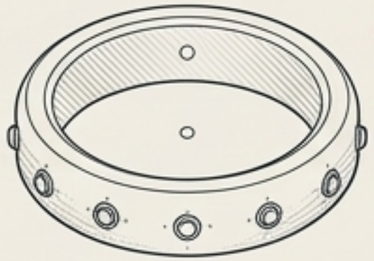
1. A node announces the Fano line (1-7) it is exploring.
2. The receiver validates the timestamp and the HD Wallet path.
3. Nodes never exchange raw matrices. The unique fingerprint of a Fano line (the change in sound-light ratio) remains strictly local and private.

Outcome: Lightweight, globally verifiable trust without centralized surveillance.

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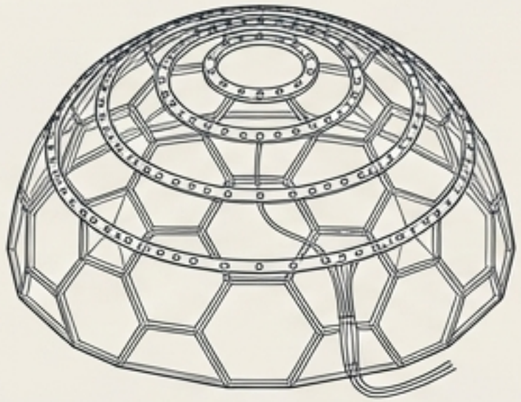
Physical Projection: Manifesting the Math

7-bit Talisman



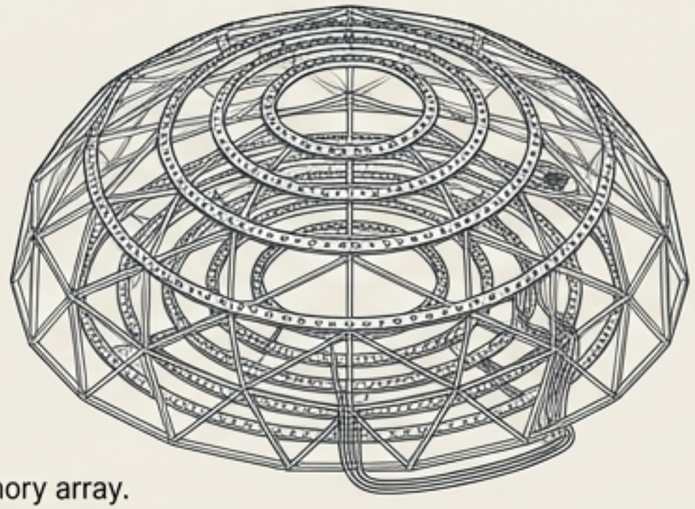
Wearable ring representing the 7 points + observer.

61-bit Personal Dome



5-ring hexagonal geodesic ESP32 array.

241-bit Garden



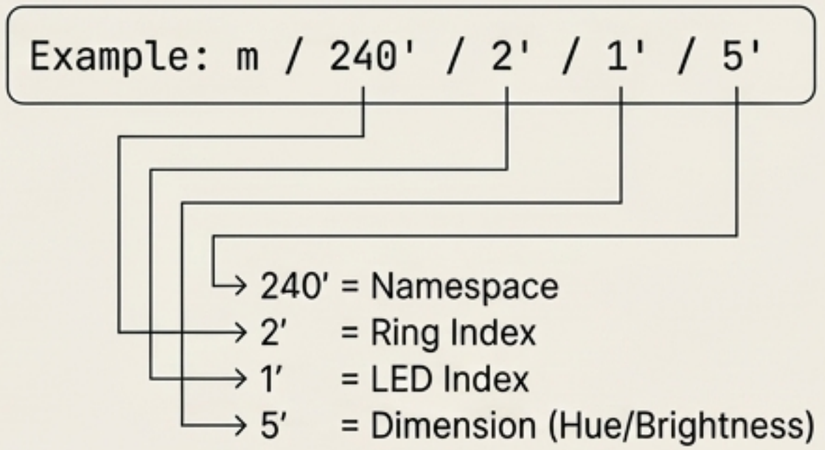
9-ring collective memory array.

Hardware Tiers:

- 7-bit Talisman: Wearable ring representing the 7 points + observer.
- 61-bit Personal Dome: 5-ring hexagonal geodesic ESP32 array.
- 241-bit Garden: 9-ring collective memory array.

HD Wallet Addressing:

Borrowed from BIP32 structure to precisely trace provenance in physical space.



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Protocol in Action: The Wisdom Fractal

The Scenario

Inputting the single word “wisdom” into the Light Garden terminal.

The Trace

- Word tokenized and mapped via WordNet.
- Anchored deterministically to Fano Line 4.
- Rotated sequentially through the Epistemic Square.

The Output

The seed word blooms into a 347-node geometric knowledge cloud, deterministically rendered in exactly 77 NDJS0N trace frames (7.7 seconds).

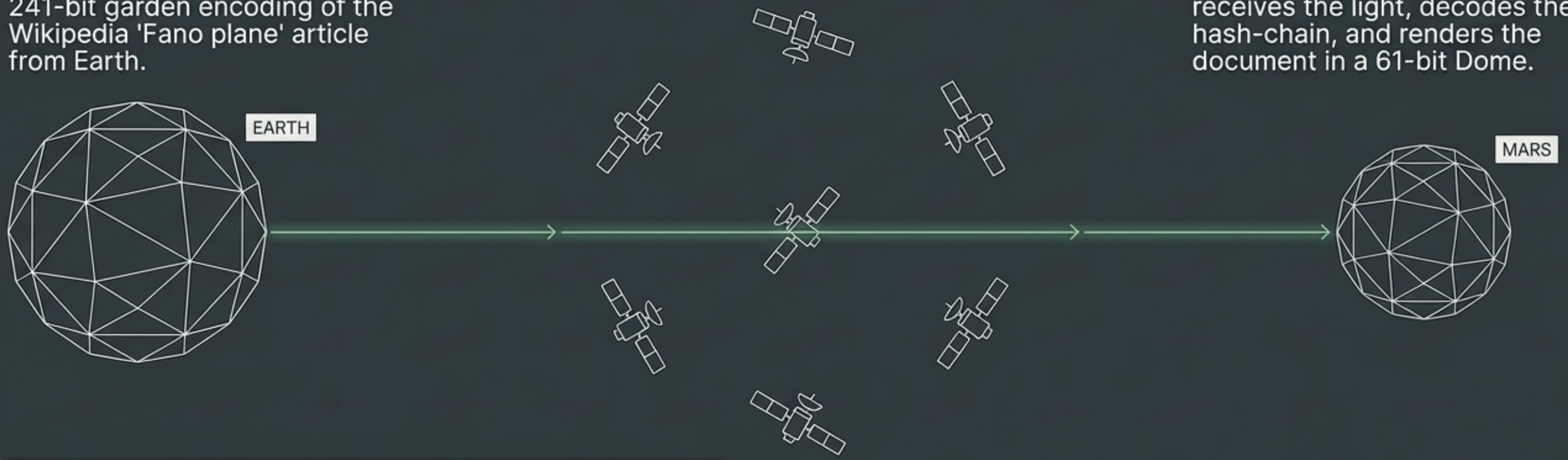


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The Interplanetary Demo

The Transmission: Sending the 241-bit garden encoding of the Wikipedia 'Fano plane' article from Earth.

The Result: Mars Base Alpha receives the light, decodes the hash-chain, and renders the document in a 61-bit Dome.



Technical Specifications

- **Encoding:** 24-bit BIP39 seed (7x2 matrix + 10-bit angle).
- **Transport:** 100W @ 532nm laser relay via 7-satellite constellation.
- **Distance & Time:** 225,000,000 km in 12.5 minutes at 100 Gbps.

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The 8th Point is You.



The Fano plane has 7 points. The 8th point—the observer, the question-asker, the one who reads—is implied by the circle that passes through points 4, 7, and 2. The gate only opens when interacted with.

Every $3 ESP32 is a seed.
Every $0.50 LED is a flower.

Join the federation.
Compile the Haskell v1 scaffold, build a dome, and share a line.

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